Objective: Build a Sales & Profit Report for Tailwind Traders

Project Showcases: Data Overview → Data Extraction → Transformation & Modeling → DAX → visualizations

Data Overview (First 10 Rows):

1. Fact Sales:

Customer Name	CustomerID	Address	Zip Code	Phone	Email	Loyalty Points
Jane Smith	C024	121 Blvd	46575	1112222	jsmith@mail.com	7
Darren White	C025	45 Royal Ln	B2M 3N5	2223333	dwhite@mail.com	25
Lucy Roberts	C026	23 Paris St	75008	3334444	lucyR@mail.com	30
Mark Lewis	C027	67 Berlin Rd	10178	4445555	markL@mail.com	20
Olivia Graham	C028	99 Madrid Ln	28014	5556666	oliviaG@mail.com	10
ack Thompson	C029	45 Rome Blvd	185	6667777	jackT@mail.com	15
Mia Hughes	C030	11 Delhi Ave	110001	7778888	miaH@mail.com	5
Aaron Walker	C031	23 Tokyo St	100-0004	8889999	aaronW@mail.com	20
Lily Peterson	C032	33 Sydney Rd	2000	9990000	lilyP@mail.com	3
Ben Foster	C033	56 Ontario St	K1A 0A6	1497911	benF@mail.com	12

Details about Customers

Product Name	Product Description	Product Category	SKU Number	Weight	Color	Size	Rating	Stock
UltraGrip Hammer	A sturdy hammer with an ergonomic grip for carpentry works.	Tools	TWH-024	0,1	Red	Small	4.5	320
Luminous Bulb 60W	Energy-saving 60W bulb with a bright luminous output.	Lighting	TWF-025	0,2	Blue	Medium	4.8	150
Oakwood Shelf	A durable oakwood shelf for home and office use.	Furniture	TWK-026	2,5	Green	Large	5.0	85
ProTile Cutter	High precision tile cutter for professional use.	Tools	TWF-027	8,0	Red	XL	4.4	190
Garden Glove Set	Comfortable gloves for gardening tasks.	Gardening	TWT-028	1,2	Black	Small	4.9	310
Steel Nails (1inch)	Rust-resistant steel nails for construction.	Hardware	TWH-029	1,5	White	Medium	4.2	210
Luxury Paint (Blue)	Premium quality blue paint for interior walls.	Paint & Deco	TWS-030	0,8	Red	Large	4.7	255
Kitchen Sink (Steel)	Stainless steel kitchen sink with a modern design.	Plumbing	TWF-031	0,1	Blue	Small	4.6	265
Power Drill Set	A complete drill set with multiple bits for various tasks.	Tools	TWG-032	0,5	Green	XL	4.0	330
Floral Wallpaper	Beautiful floral design wallpaper for home decor.	Paint & Deco	TWT-033	0,2	Yellow	Small	4.8	500

Details about Products

OrderID	Gross Product Price	Tax Per Product	Quantity Purchased	Country ID	Sales Rep
1024	60	8.4	2	1	Alice
1025	35	2.45	3	5	Bob
1026	250	17.5	1	1	Alice
1027	15	1.05	5	2	Charlie
1028	55	3.85	4	1	Alice
1029	75	5.25	2	2	David
1030	50	14.7	1	3	Charlie
1031	90	6.3	3	2	Alice
1032	180	12.6	2	1	Bob
1033	3	1.4	6	1	David

Sales Data

2. Dim Purchase:

PurchaseID	Supplier	Last Visited	ReturnStatus	Warranty (Months)	Purchase Date	Return Policy (Days)	Feedback	OrderID
1	Smith Tools Inc.	10/10/2023	Not Returned	6	09/20/2023	30	Wont Recommend	1024
2	BrightLife Co.	10/12/2023	Returned	12	09/22/2023	45	Neutral	1025
3	TimberCraft Suppliers	10/11/2023	Not Returned	24	09/23/2023	60	Will Recommend	1026
4	BuildMaster Tools	10/14/2023	Returned	36	09/25/2023	30	Neutral	1027
5	GreenThumb Essentials	10/15/2023	Not Returned	18	09/26/2023	15	Will Recommend	1028
6	BuildRight Supplies	10/16/2023	Not Returned	12	09/27/2023	20	Wont Recommend	1029
7	ColorDreams Paints	10/14/2023	Not Returned	6	09/28/2023	25	Neutral	1030
8	AquaFlow Suppliers	10/15/2023	Returned	24	09/29/2023	30	Wont Recommend	1031
9	ToolTech Incorporated	10/16/2023	Not Returned	6	09/30/2023	20	Will Recommend	1032
10	DecorDelight Co.	10/17/2023	Not Returned	12	10/01/2023	40	Neutral	1033

- The Sales table currently contains 22 columns. To optimize performance and reduce redundancy, descriptive attributes are normalized into two dimension tables Dim Customers and Dim Products. Transactional data remains in the Fact Sales table.

3. Dim Countries:

Country ID	Country	Exchange ID
1	USA	1
2	UK	2
3	France	3
4	UAE	4
5	Australia	5

4. Exchange Rates:

Exchange ID	ExchangeRate	Exchange Currency
1	1	USD
2	0.75	GBP
3	0.85	EUR
4	3.67	AED
5	1.3	AUD

- Exchange Rates can be merged with Dim Countries using Exchange ID as the primary key. This merge simplifies the model by reducing redundant dimensions and promotes a cleaner star schema design.

Data Extraction using SQL

1st Query Fact Sales:

```
select *,(T.[Gross Revenue] - T.[Total Tax]) as 'Net Revenue' ,
(T.[Gross Revenue] - T.[Total Tax]) - ([Cost Per Unit]*Quantity_Purchased) as 'Profit'
from
    (select(Gross_Product_Price*0.35) as 'Cost Per Unit',
    (Gross_Product_Price*Quantity_Purchased) as 'Gross Revenue',
    (Tax_Per_Product*Quantity_Purchased) as 'Total Tax',
    OrderID,Gross_Product_Price,Tax_Per_Product,
    Quantity_Purchased,SKU_Number,Country_ID,CustomerID
    from tblFactSales) as T
```

References Sub-Query that Calculates Net Revenue & Profit to generate financial insights.

Sub-Query

2nd Query Dim Customer:

```
with Dimcustomers as
(Select CustomerID,Customer_Name,Address,Zip_Code,Phone,Email,Loyalty_Points
 from tblFactSales)
Select * from Dimcustomers
```

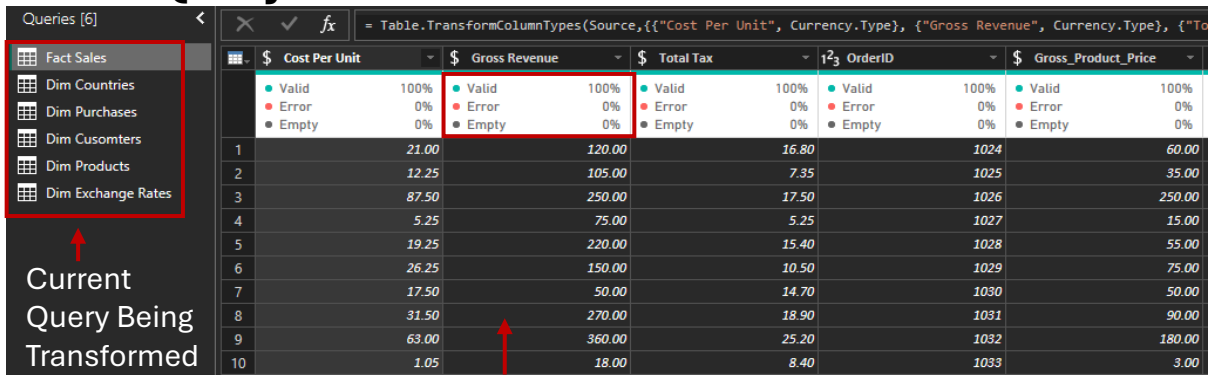
- CTEs help normalize Fact table and prepare clean dimensions.

3rd Query Dim Products:

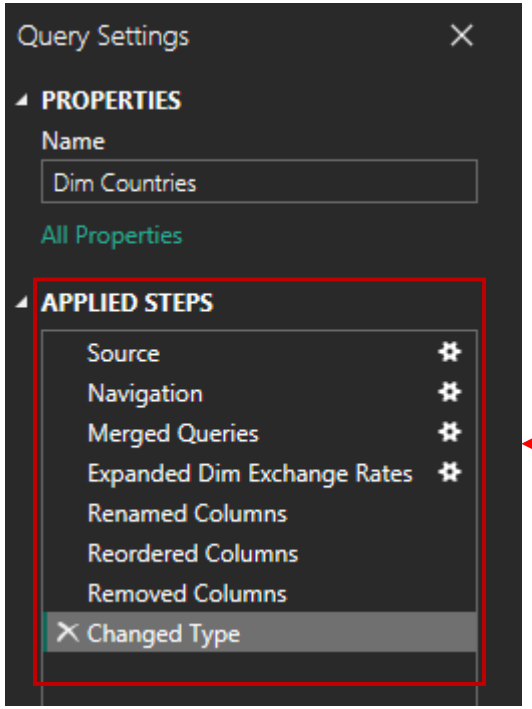
```
use [Tailwind Traders ]
go
with DimProduct as
(Select Product_Name,Product_Description,Product_Category,SKU_Number,Weight,Color,Size,Rating,Stock
    from tblFactSales)
Select * from DimProduct
```

Data Transformation & Modeling

Power Query:



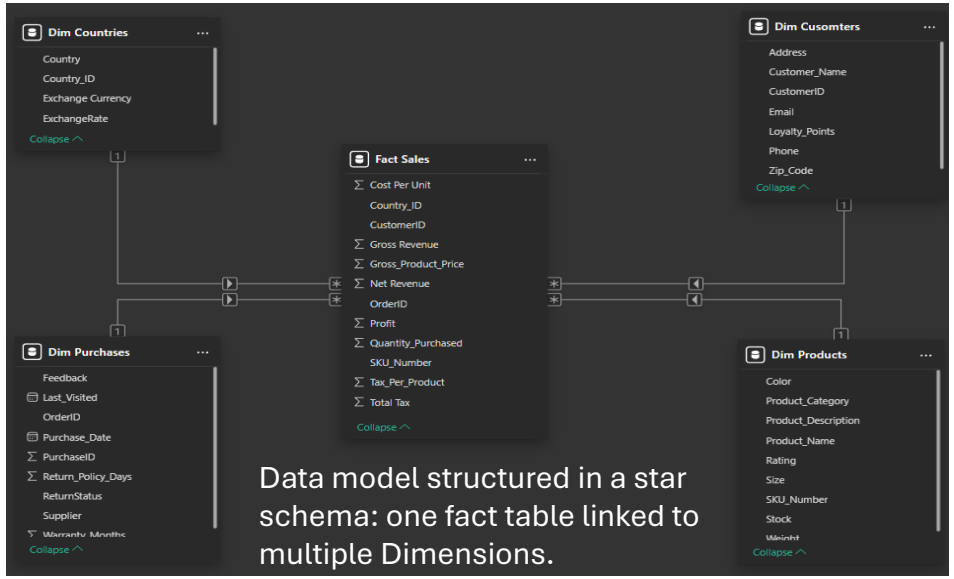
Power Query transformations ensure data integrity, all columns validated with zero nulls or errors.



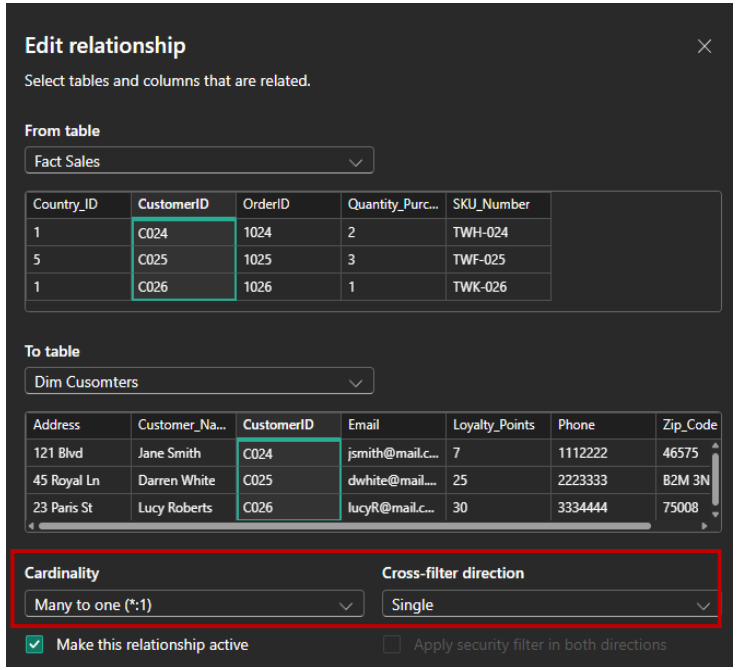
Key transformations include:

- data type corrections
- removal of redundant columns
- normalization of dimensions through merge.

Data Model:



Data model structured in a star schema: one fact table linked to multiple Dimensions.

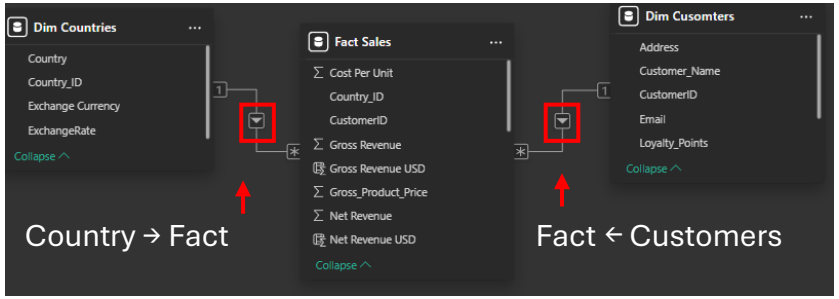


Relationships follow Many-to-One cardinality with Single cross-filter direction to optimize performance.

DAX

Certain performance measures like aggregating Customer Loyalty Points by Country, require reversing the default single-direction filter flow.

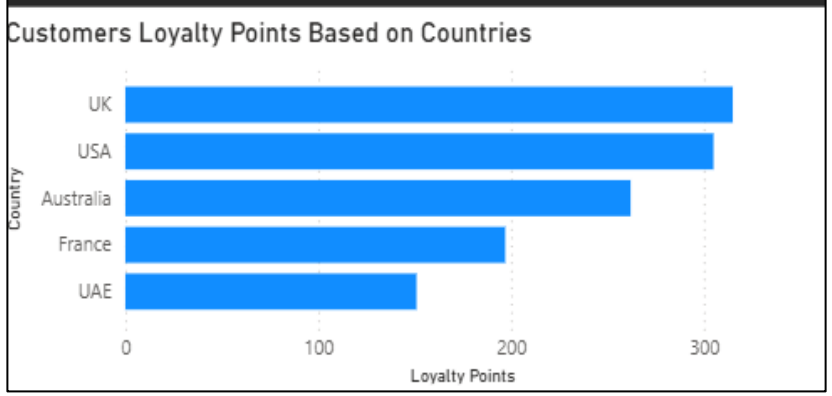
Current Filter Direction in the model:



To compute metrics that instead flow from Fact → Customers, we use the Cross-filter function to temporarily override the relationship direction.

Dax and the Visual it's used in:

```
Loyalty Points =
CALCULATE(SUMX('Dim Cusomters','Dim Cusomters'[Loyalty_Points]),
CROSSFILTER('Dim Cusomters'[CustomerID],'Fact Sales'[CustomerID],Both))
```



To support Time-Intelligence calculations (YTD, QTD, MTD), a dedicated date table is created through Dax and marked as a Date table in the model.

```
DateTable =
ADDCOLUMNS
(CALENDAR(DATE(2020, 1, 1), DATE(2023, 12, 31)),
"Year", YEAR([Date]),
"Month Number", MONTH([Date]),
"Month", FORMAT([Date], "MMMM"),
"Quarter", QUARTER([Date]),
"Weekday", WEEKDAY([Date]),
"Day", DAY([Date]))
```

Following Dax utilizes date table to create metrics for Profit Overview report

```
Yearly Profit Margin = DIVIDE(SUM('Fact Sales'[Profit USD]),SUM('Fact Sales'[Net Revenue USD]))
```

```
Quarterly profit Margin =
CALCULATE([Yearly Profit Margin],DATESQTD(DateTable[Date]))
```

```
YTD Profit Margin = CALCULATE([Yearly Profit Margin],DATESYTD(DateTable[Date]) )
```

Uses Exchange rate to convert relevant metrics for Sales overview report

```
Gross Revenue USD = 'Fact Sales'[Gross Revenue] * RELATED('Dim Countries'[ExchangeRate])
```

```
Net Revenue USD = 'Fact Sales'[Net Revenue] * RELATED('Dim Countries'[ExchangeRate])
```

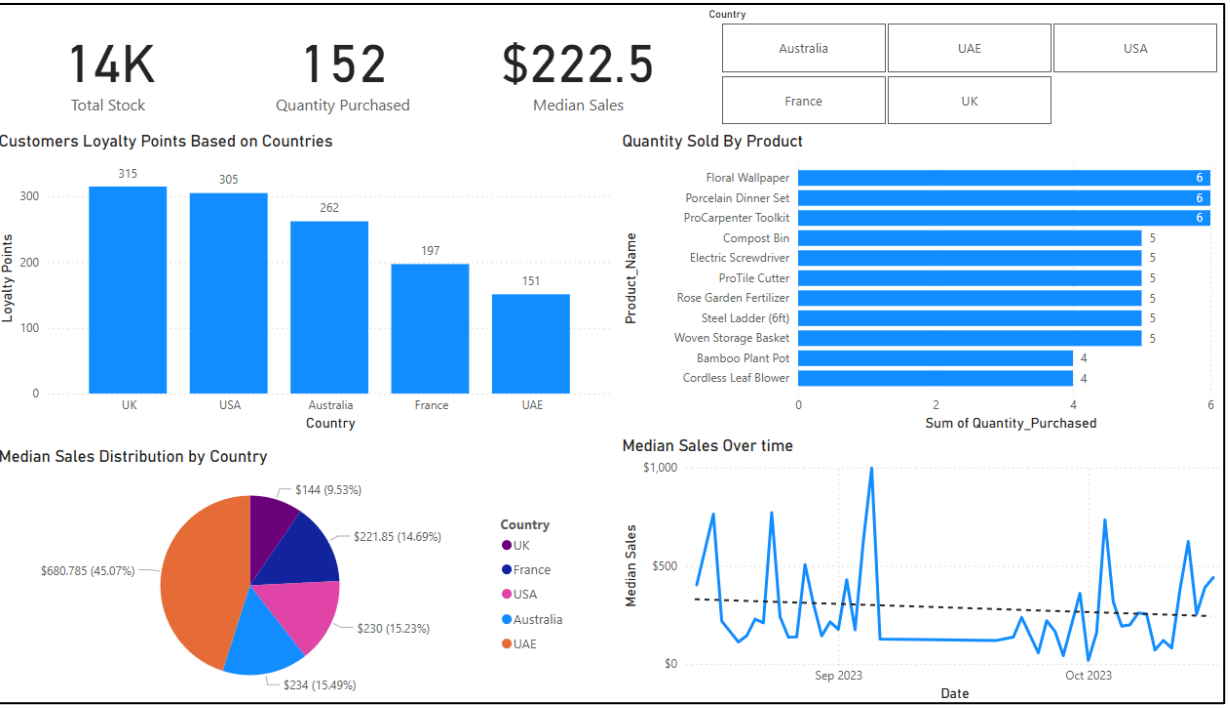
```
Profit USD = 'Fact Sales'[Profit] * RELATED('Dim Countries'[ExchangeRate])
```

Median being unaffected by outliers can be an effective metric to measure sales performance overtime through Line-chart

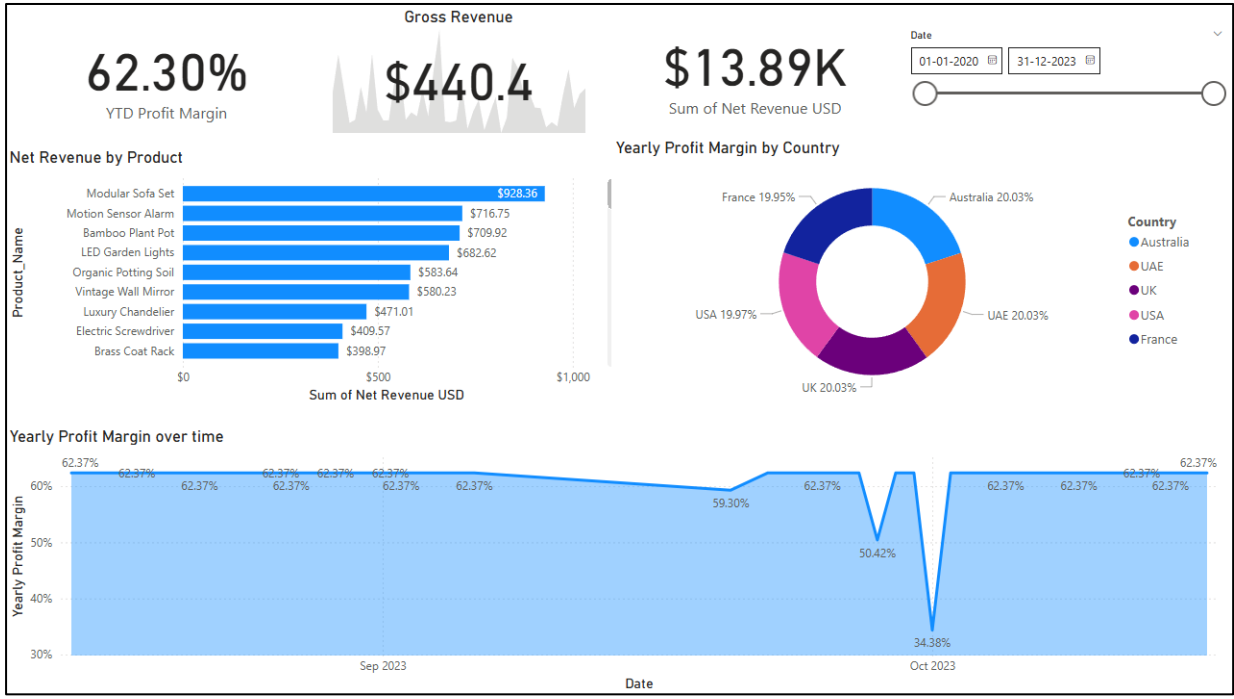
```
Median Sales = MEDIAN('Fact Sales'[Gross Revenue USD])
```

Visualization Sales & Profit Insights

Sales overview report



Profit overview report



The Report consolidates key performance indicators for Sales and Profit, enabling a quick understanding of organizational performance across time, product categories, and countries.
Although the current dataset uses sample data, each visual is designed to demonstrate analytical depth and decision-making value for example:

In Sales report:	In Profit Report:
Customer's Loyalty in UAE is lowest but if we look at the median sales of that country, its contribution of 45.07% is highest compared to others. This tell us although we are doing great sales in that country, customer retention towards our products is not high meaning most of our sales are From new customers	The Donut chart showcases the impact of exchange rates on profit margins. Each country maintains a similar margin (±0.10%) implying that to sustain consistent product pricing across markets, costs are being managed efficiently resulting in stable profit margins despite currency differences.